

probe libc's `inet_pton` & backtrace it with `ping` : failed!

To probe and backtrace libc's `inet_pton` function using `ping`, you must execute the tracing utility with root privileges (`sudo`) because kernel probes (`kprobes`) and user probes (`uprobes`) require elevated administrative permissions.

Here is how to successfully trace and backtrace the function using `bpfttrace` or `ftrace`.

Solution 1: Using bpfttrace (Recommended)

`bpfttrace` is the most efficient tool for tracking user-space functions and printing their stack traces.

1. Install bpfttrace

bash

```
sudo apt install bpfttrace # Debian/Ubuntu
```

```
sudo dnf install bpfttrace # Fedora/RHEL
```

Use code with caution.

2. Execute the probe

Run this command in one terminal, then run `ping 8.8.8.8` in another:

bash

```
sudo bpfttrace -e 'uprobe:/lib/x86_64-linux-gnu/libc.so.6:inet_pton
{ printf("pid: %d, comm: %s\n", pid, comm); print(ustack); }'
```

Use code with caution.

Note: Path to `libc.so.6` may vary. Find yours using `ldd $(which ping)`.

Solution 2: Using ftrace (Built-in Linux Tool)

If you cannot install third-party tools, use the built-in kernel tracing infrastructure.

1. Navigate to the trace directory

bash

```
cd /sys/kernel/debug/tracing/
```

Use code with caution.

2. Define the uprobe

bash

```
echo 'p:/lib/x86_64-linux-gnu/libc.so.6:inet_pton' > uprobe_events
```

Use code with caution.

3. Enable the trigger and stack trace

bash

```
echo 1 > events/uprobes/p_libc_so_6_inet_pton/enable
```

```
echo 'stacktrace' > events/uprobes/p_libc_so_6_inet_pton/trigger
```

Use code with caution.

4. View the output

bash

cat trace

Use code with caution.

[1]

Common Failure Reasons

- **Missing Sudo:** Forgetting `sudo` will cause immediate initialization failure.
- **Incorrect libc Path:** The path to `libc.so.6` changes depending on your Linux distribution architecture.
- **Stripped Symbols:** If your `libc` lacks a symbol table, debug symbols (`libc6-dbg`) must be installed.
- **Modern Ping Behavior:** Some modern versions of `ping` use `getaddrinfo` instead of calling `inet_pton` directly.